5.3 Thermodynamics

This topic covers specific heat capacity, specific latent heat, internal energy and the gas equation.

This topic may be studied using applications that relate, for example, to space technology.

This unit includes many opportunities for developing experimental skills and techniques by carrying out more than just the core practical experiments.

Candidates will be assessed on their ability to:

125	be able to use the equations $\Delta E = mc\Delta\theta$ and $\Delta E = L\Delta m$
126	**CORE PRACTICAL 12: Calibrate a thermistor in a potential divider circuit as a thermostat**
127	**CORE PRACTICAL 13: Determine the specific latent heat of a phase change**
128	understand the concept of *internal energy* as the random distribution of potential and kinetic energy amongst molecules
129	understand the concept of *absolute zero* and how the average kinetic energy of molecules is related to the absolute temperature
130	be able to use the equation $pV = N\mathrm{k}T$ for an ideal gas
131	**CORE PRACTICAL 14: Investigate the relationship between pressure and volume of a gas at fixed temperature**
132	be able to derive and use the equation $\frac{1}{2}m\langle c^2\rangle = \frac{3}{2}\mathrm{k}T$